Fiscal Year:	FY 2020	**Task Last Updated:**	FY 08/17/2020
PI Name:	Axelbaum, Richard Ph.D.		
Project Title:	Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (80NSSC20M0072 ; NNX15AC75A)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	rla@wustl.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	314 935-7560
Organization Name:	Washington University		
PI Address 1:	Department of Energy, Environmental and Chemical Engineering		
PI Address 2:	305 Jolley Hall		
PI Web Page:			
City:	St. Louis	**State:**	MO
Zip Code:	63130	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	12/12/2014	**End Date:**	04/30/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Stocker, Dennis P	**Contact Phone:**	216-433-2166
Contact Email:	dennis.p.stocker@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 4/30/2022 with new grant number 80NSSC20M0072, per NSSC information (Ed., 8/12/20) NOTE: Extended to 06/11/2020 per NSSC information (Ed., 1/29/2020) NOTE: Extended to 12/11/2019 per NSSC information (Ed., 2/3/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Sunderland, Peter Ph.D. (University of Maryland) Urban, David Ph.D. (NASA Glenn Research Center)		
Grant/Contract No.:	80NSSC20M0072 ; NNX15AC75A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE Update August 2020: New grant issued for Flame Design - A Novel Approach to Clean Efficient Diffusion Flames, with grant number 80NSSC20M0072, which continues to 4/30/2022. NOTE (September 2015): Continuation of Flame Design - A Novel Approach to Clean Efficient Diffusion Flames (NNX09AO53A). The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the PI (Principal Investigator), the Co-Is (Co-Investigators), and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, and Peter Sunderland directing experiments and numerical simulations at University of Maryland.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential for improving efficiency and reducing pollutant emissions in terrestrial combustors. For example, it is contributing to novel designs of future coal power plants.
Task Progress:	We have completed Round 1 of Flame Design testing and are processing the data. This includes making videos from HiBMs still images, converting video from the ops camera from ts to mp4 format, plotting the digital data for all tests, and measuring flame diameters. We have finalized the test matrix for Round 2. We are organizing the results of Flame Design Round 1 (flame diameter, fiber temperature, PMTs, radiometers, etc.). Particular focus has been placed on using two-color ratio and ratio intensity pyrometry to obtain fiber temperature. We continued to plot and analyze the thermocouple data. We are processing HiBMs data to obtain soot volume fraction and are using the Advanced Combustion via Microgravity Experiments (ACME) images to obtain soot temperature. Fiber temperature is being used to measure flame temperature at the sooting limit. We have compiled all of our radiative extinction data, excluding extinction temperature. We have performed thin-filament pyrometry for several International Space Station (ISS) flames based on color ratios and based on grayscale intensity. We have completed soot temperature measurements of ethylene and methane tri-axial flames using two-color pyrometry. We performed narrow-angle radiometer measurements of radiant emissions in ethylene tri-axial flames. Soot volume fraction measurements using color pyrometry and an absolute intensity calibration have been performed. In our research on double luminous zones, we have made our chemiluminescence images quantitative and have chosen some of them to be displayed as color contour plots of intensity at specific wavelengths. Thermocouple probing of the double luminous zones was performed. Work continued on the imaging of excited species. An analytical model of steady-state diameters of gas-fed spherical diffusion flames was created. Our work on the numerical model has continued. We have run multiple cases with the University of California San Diego (UCSD) mechanism. We continue to investigate the boundary conditions for radiation and conduction. Because burner heating can significantly affect the flame diameter and lifetime, we have started to implement a heat balance at the burner surface to solve for burner temperature and better conserve enthalpy. We simulated ethylene flames to perform parametric studies in an attempt to find conditions that are favorable to steady state flames. We compared the simulation results with experimental results. Work is underway to update the globally implicit solver to the time-splitting method. We modified the SphDiff code to plot the radius of stoichiometric mixture fraction as a function of time, both for non-reacting and reacting flows. We added 2nd order upwind scheme. We modified the transient term in the conservation of mass equation so that it captures the density change due to temperature and concentration changes. We added an automatic temperature initialization scheme from an analytical droplet combustion model. 2. Scientific Context, Hypothesis, and Research Objectives: The overwhelming majority of practical combustion processes involve non-premixed combustion of hydrocarbons. These processes must be efficient and the resulting emissions low. For example, soot production, which can adversely affect efficiency, emissions, and equipment lifetime, must be controlled, and flame extinction must be avoided. Many methods have been considered to reduce soot, such as increasing strain rates and inert concentrations and reducing pressures, but these techniques generally weaken the flame, leading to extinction and resulting in substantial performance penalties. The proposed study considers a novel method of approaching combustion to address these concerns. The approach is termed “Flame Design” as it attempts to take our fundamental understanding of diffusion flames and, within the inherent constraints imposed on the problem, design flames that best optimize efficiency and minimize pollutants. This effort is a collaborative effort between the Principal Investigator (PI), the Co-Is, and staff at NASA Glenn Research Center. The division of labor has Richard Axelbaum directing normal gravity experiments and computations at Washington University, David Urban providing guidance for the flight experiments, and Peter Sunderland directing experiments and numerical simulations at University of Maryland.
Bibliography Type:	Description: (Last Updated: 11/26/2025)
Abstracts for Journals and Proceedings	Irace P, Lee HJ, Waddell K, Tan L, Stocker DP, Urban DL, Sunderland PB, Axelbaum RL. "Steady-State and Radiatively Extinguishing Spherical Diffusion Flames in Microgravity." 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. Abstracts. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019

Abstracts for Journals and Proceedings	Irace P, Tan L, Waddell K, Lee HJ, Stocker DP, Urban DL, Sunderland PB, Axelbaum RL. "A Galaxy of Soot Particles." Art Competition Entry, 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. 35th Annual Meeting of the American Society for Gravitational and Space Research, Denver, CO, November 20-23, 2019. , Nov-2019
Articles in Peer-reviewed Journals	Wang Z, Sunderland PB, Axelbaum RL. "Dilution effects on laminar jet diffusion flame lengths." Proceedings of the Combustion Institute. 2019 Jan;37(2):1547-53. https://doi.org/10.1016/j.proci.2018.06.085 , Jan-2019
Articles in Peer-reviewed Journals	Wu W, Adeosun A, Axelbaum RL. "A new method of flame temperature measurement utilizing the acoustic emissions from laser-induced plasmas." Proceedings of the Combustion Institute. 2019 Jan;37(2):1409-15. https://doi.org/10.1016/j.proci.2018.07.096 , Jan-2019
Papers from Meeting Proceedings	Irace PH, Yang Z, Gopan A, Axelbaum RL. "Soot Formation and Radiation Heat Transfer in Tri-axial Diffusion Flames." 11th U.S. National Combustion Meeting, Pasadena, CA, March 24-27, 2019. Meeting papers.11th U.S. National Combustion Meeting, Pasadena, CA, March 24-27, 2019. , Mar-2019
Papers from Meeting Proceedings	Wang Z, Sunderland PB, Axelbaum RL. "Double Luminous Zones in Laminar Jet Diffusion Flames." 11th U.S. National Combustion Meeting, Pasadena, CA, March 24-27, 2019. Meeting papers.11th U.S. National Combustion Meeting, Pasadena, CA, March 24-27, 2019. 9 p. , Mar-2019